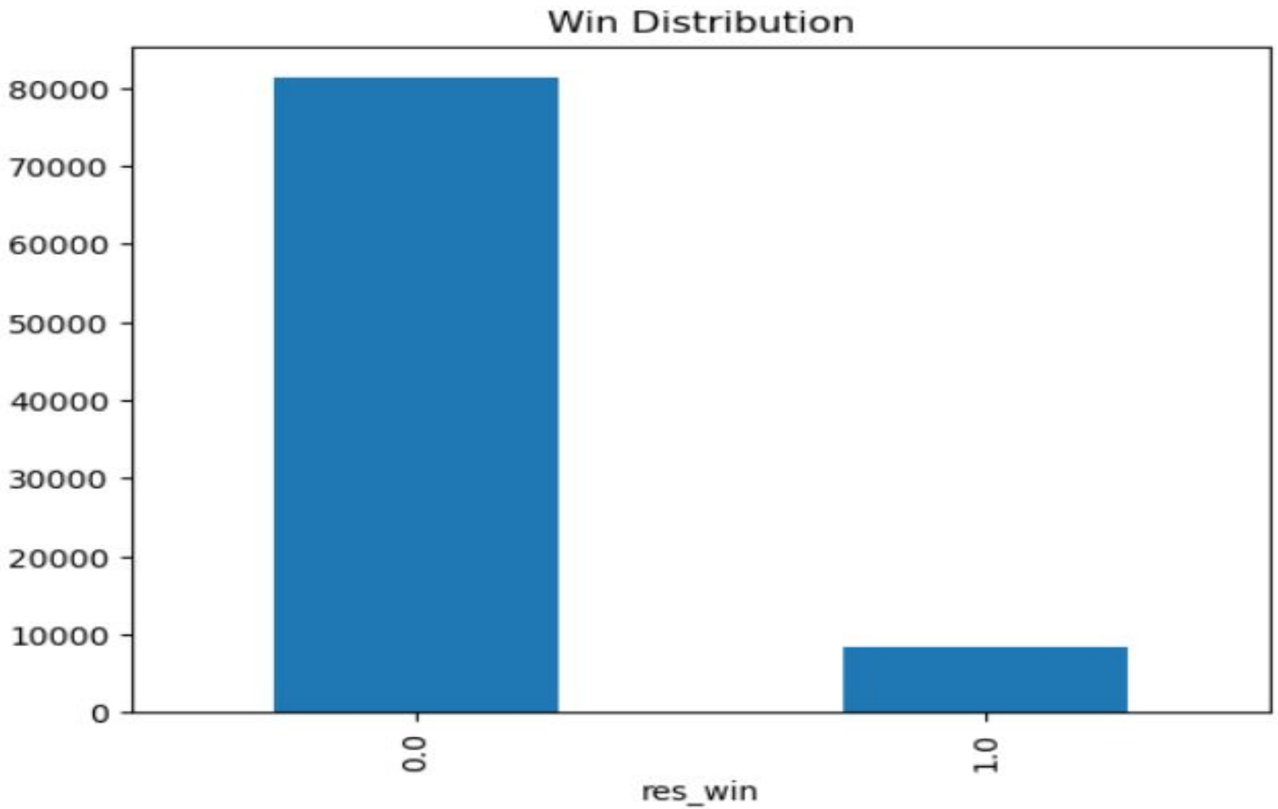
Exploratory Data Analysis

[5]:
```
df['res_win'].value_counts().plot(kind='bar')
plt.title('Win Distribution')
plt.show()
```



Build Prediction Model

[9]:
```
X = df[features]
y = df[target]
cat_cols = X.select_dtypes(include='object').columns
num_cols = X.select_dtypes(exclude='object').columns

preprocessor = ColumnTransformer([
 ('num', SimpleImputer(strategy='median'), num_cols),
 ('cat', Pipeline([('imp', SimpleImputer(strategy='most_frequent')),
                   ('ohe', OneHotEncoder(handle_unknown='ignore'))]), cat_cols)
])

model = Pipeline([
 ('prep', preprocessor),
 ('rf', RandomForestClassifier(n_estimators=200, random_state=42))
])

X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)
model.fit(X_train,y_train)
pred = model.predict(X_test)
prob = model.predict_proba(X_test)[:,1]

print('Accuracy:', accuracy_score(y_test,pred))
```

```
Accuracy: 0.9049556831484475
```

Probability & Fair Odds

[10]:
```
results = X_test.copy()
results['win_probability'] = prob
results['fair_odds'] = 1 / np.clip(prob,0.01,None)
results[['win_probability','fair_odds']].head()
```

[10]:

	win_probability	fair_odds
85069	0.040	25.000000
24385	0.150	6.666667
19869	0.030	33.333333
81368	0.135	7.407407
33964	0.015	66.666667

ROI Backtesting

[11]:

```
results['bet'] = (results['win_probability'] > 0.5).astype(int)
results['stake'] = results['bet']
results['profit'] = np.where(results['bet']==1, results.get('decimalPrice',1), 0)
roi = (results['profit'].sum()-results['stake'].sum()) / max(results['stake'].sum(),1)
print('ROI:', roi)
```

```
ROI: -0.48301850067625085
```

AI Race Explanation

[12]:

```
def explain(horse, prob):
    return f'{horse} has a {prob:.1%} predicted win probability based on age, odds, trainer, jockey and historical race signals.'

print(explain('Strong Suspicion',0.72))
```

```
Strong Suspicion has a 72.0% predicted win probability based on age, odds, trainer, jockey and historical race signals.
```